

LeetCode 73: Set Matrix Zeroes

Authored by: Arpit Raj, LNMIIT Jaipur

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Target Companies

JP Morgan

TCS

Goldman Sachs

Sprinklr

Approach & Explanation

Core Intuition:

"Basically saare suspicious rows aur suspicious columns mark krte hue chalo. Last me har uss row and har uss col ko 0 se inplace replace kr do."

Complexity Analysis

TIME COMPLEXITY

$O(N \times M)$

Where N is the number of rows and M is the number of columns. We traverse the matrix fully twice.

SPACE COMPLEXITY

$O(N + M)$

Because we are using two unordered sets to store the suspicious rows and suspicious columns.

Source Code (C++)

```
class Solution {
public:
    void setZeroes(vector<vector<int>>& matrix) {

        // basically saare suspicious rows aur suspicious columns
        // mark krte hue chalo
        // last me har uss row
        // and har uss col
        // ko 0 se inplace replace kr do

        int n = matrix.size();
        int m = matrix[0].size();

        unordered_set<int> susrows;
        unordered_set<int> suscols;

        for(int i = 0; i < n; i++) {
            for(int j = 0; j < m; j++) {
                if(matrix[i][j] == 0) {
                    susrows.insert(i);
                    suscols.insert(j);
                }
            }
        }

        for(int row : susrows) {
            for(int j = 0; j < m; j++) {
                matrix[row][j] = 0;
            }
        }

        for(int col : suscols) {
            for(int i = 0; i < n; i++) {
                matrix[i][col] = 0;
            }
        }
    }
}
```

```
}  
};
```